Domain Adaptation in High-Resolution Satellite Images using Deep Learning

Mayank S. (mshetty4), Vaibhav R. (vraheja3), Viraj S. (virajns2)

CS 444: Deep Learning for Computer Vision

1 Updated Problem Definition and Plan

Originally, our project aimed to address domain adaptation issues in satellite imagery by utilizing traditional deep-learning models. However, after preliminary analysis and review of recent literature, we decided to shift our focus towards a self-supervised learning approach. This change was driven by the potential for better generalizability across different geographical locations without the need for extensive labeled datasets. The key modification in our approach involves integrating techniques from recent research, particularly the self-supervised domain-agnostic adaptation methods that have shown promise in preliminary tests, reproducing the results on different datasets and further tuning the proposed model's to improve accuracies on the new datasets.

2 Key Resources

Our implementation heavily relies on two open-source repositories which provide a foundation for our domain adaptation models:

- Domain-Adaptable Self-Supervised Representation Learning on Remote Sensing Satellite Imagery [3, 4]. This repository offers a framework for self-supervised learning that is adaptable to various domains within the context of remote sensing and satellite imagery. The approach is designed to leverage unlabeled data, which is a common scenario in satellite data where annotations can be expensive or difficult to obtain.
- Self-supervised Domain-agnostic Domain Adaptation for Satellite Images [6]. This study's approach to domain adaptation for satellite imagery does not rely on predefined domain knowledge. They propose a method, termed self-supervised domain-agnostic domain adaptation (SS(DA)^2), that utilizes a generative adversarial network with a contrastive learning component to perform image-to-image translation between any two satellite image patches.
- Unsupervised Domain Adaption for Remote Sensing Semantic Segmentation with Transformer [2]. This repository focuses on unsupervised domain adaption (UDA) for remote sensing applications. UDA enables models trained on a particular dataset (source domain) to be adapted and subsequently used on another dataset (target domain) where labels might not be available.
- The INRIA Aerial Image Labeling Dataset [1] is a collection of high-resolution aerial images used for semantic labeling. Covering 810 km^2, it's split evenly between training and testing areas. The images have a resolution of 0.3 meters, capturing urban landscapes from dense cities to small towns. It includes two labeled classes: 'building' and 'not building', with ground truth available for the training set. This dataset tests the ability of models to generalize across different urban settings and conditions.

- The WHU Building Dataset [5] is a meticulously curated collection of building samples from aerial and satellite imagery. It features over 220,000 independent buildings from aerial images with a 0.075 m spatial resolution, covering 450 km^2 in Christchurch, New Zealand1. The satellite imagery dataset is divided into two subsets: one with images from various cities worldwide and another with 6 satellite images covering 550 km^2 in East Asia with a 2.7 m ground resolution1. This dataset is ideal for testing the robustness of building extraction algorithms, given its diversity in resolution, geographic locations, and imaging conditions1.



Figure 1: Test Image and Ground Truth image from the Inria Aerial Image Labeling Dataset

We use the Inria Aerial Image Labeling Dataset [1] and the WHU Building Dataset [5] to train and test our models. Computations are performed on university-provided cloud computing resources, allowing us to leverage GPUs for training deep neural networks.

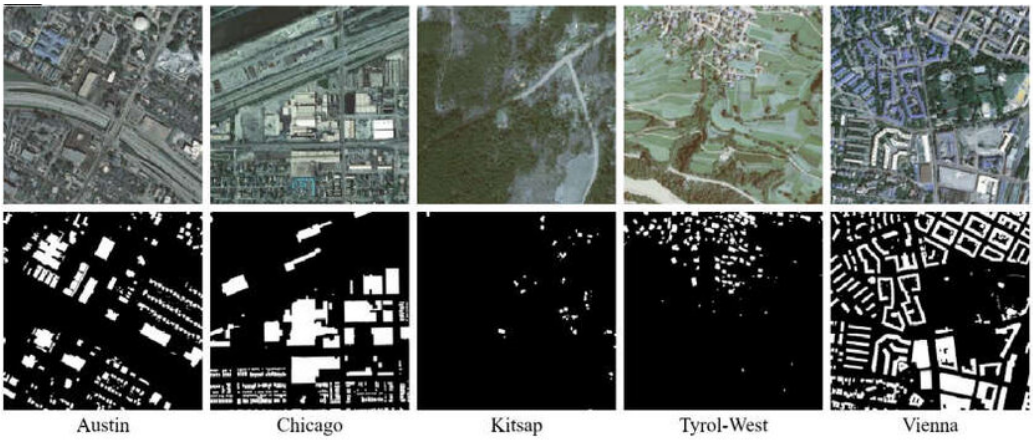


Figure 2: Test Image and Ground Truth image from the SIRI-WHU Building Dataset

3 Summary of Work Done to Date

To date, we have successfully set up the development environment and replicated the baseline models provided in the referenced GitHub repositories. Initial tests on the Inria dataset using

pre-trained models have demonstrated the feasibility of adapting these models to our specific tasks. As proposed by Zhang et al. [6] we have followed an Image-to-Image network framework. The framework follows an encoder and decoder architecture consisting of a 2D convolution layer, an instance normalization layer, a max pooling layer, and a ReLU activation layer. The decoders followed a Unet architecture for generation. The downstream semantic segmentation model uses a ResNet50 backbone. We have started experimenting with modifying the training process to suit our datasets better as we are not using the same dataset originally used by the others. We are also looking at tuning the hyperparameters of learning rate, decay, and number of epochs to understand the model's performance better. We have looked at modifying the architecture of the model to further improve the model and are looking at ensembling the model's architecture to architectures we have seen in other implementations [3].

4 Plan for the Remainder of the Project

In the coming weeks, our plan is to:

- Improve accuracy and reduce overfitting by fine-tuning model parameters.
- Evaluate the models on the Inria and WHU datasets to compare performance and identify potential areas for improvement.

Risks include potential model overfitting and underperformance on the datasets. Our mitigation strategies include using validation during training and trying to implement ensemble methods to improve generalization.

5 Member Roles

- **Mayank S.** performs data preprocessing and augmentation, providing training and testing data from selected datasets.
- **Vaibhav R.** is mainly concerned with training models, adjusting hyperparameters, and tuning the models o based on datasets and inputs provided by teammates.
- **Viraj S.** is responsible for the evaluation of the model and helps in the documentation of code and hyperparameters for tuning the models to provide sufficient analysis and progress updates.

Going forward, we plan weekly meetings to review progress, discuss challenges, and redistribute tasks as needed to ensure the project remains on track. We have worked in close collaboration with each other during the past weeks and everyone, irrespective of roles assigned, has provided great support in research and understanding of the topic, the challenges we might face, and coming up with interesting solutions. Throughout the process, we have handled various problems encountered and have come to a consensus on how to solve the domain adaptation task.

References

[1] *Inria aerial image labeling dataset*, `https://project.inria.fr/aerialimagelabeling/files/`, Accessed: 2021-09-20.

[2] Levantespot, *Uda for rs*, `https://github.com/Levantespot/UDA_for_RS`, 2021.

[3] muskaan712, *Domain-adaptable self-supervised representation learning on remote sensing satellite imagery*, `https://github.com/muskaan712/Domain-Adaptable-Self-Supervised-Representation-Learning-on-Remote-Sensing-Satellite-Imagery`, 2021.

[4] T. Postadjian, A. Le Bris, H. Sahbi, and C. Mallet, "Domain adaptation for large scale classification of very high resolution satellite images with deep convolutional neural networks," in *IGARSS 2018*, 2018.

[5] *Whu building dataset*, `http://www.lmars.whu.edu.cn/prof_web/zhongyanfei/e-code.html`, Accessed: 2021-09-20.

[6] F. Zhang, Y. Shi, and X. X. Zhu, "Self-supervised domain-agnostic domain adaptation for satellite images," *Journal of LaTeX Class Files*, vol. 14, no. 8, 2023.